

SAMPLE LESSON CONTENT

Properties of Matter: Solid, Liquid, and Gas

Everything around us is made of matter. Matter is anything that takes up space and has mass, meaning it can be weighed. The chair you sit on, the water you drink, and even the air you breathe are all forms of matter, even though they look and behave very differently from one another.

Matter is commonly found in three states: solid, liquid, and gas. Each state has its own set of properties that can be observed without needing any special equipment, simply by looking at, touching, or pouring the substance in question.

Solids

A solid is matter that has both a definite shape and a definite volume. This means a solid keeps its shape and size no matter what container it is placed in, unless an outside force is applied to it. A stone, a book, a wooden block, and an ice cube are all examples of solids.

The particles that make up a solid are packed very closely together in a fixed arrangement. Because these particles can only vibrate in place and cannot move past one another, solids do not flow and are generally difficult to compress.

Liquids

A liquid has a definite volume but no definite shape of its own. Instead, a liquid takes the shape of whatever container it is poured into, while the amount of space it occupies stays the same. Water, cooking oil, milk, and juice are common examples of liquids.

In a liquid, particles are still close to one another, but they are able to slide and move past each other. This is what allows liquids to flow and to be poured from one container to another.

Gases

A gas has neither a definite shape nor a definite volume. A gas will expand to completely fill whatever space or container it is placed in, no matter how large that space is. The air inside a balloon, the steam rising from a hot cup of soup, and the oxygen in the atmosphere are all gases.

The particles in a gas are spread far apart from one another and move quickly and freely in all directions. Because there is so much empty space between gas particles, gases can be compressed much more easily than solids or liquids.

Comparing the Three States

The main difference among solids, liquids, and gases lies in how closely their particles are packed together and how freely those particles can move. Solids hold their shape because their particles

are locked in place; liquids flow because their particles can slide past each other while still staying close together; and gases spread out because their particles move freely with large gaps between them.

Some materials, such as water, can exist in all three states depending on their temperature — as ice (solid), as liquid water, or as water vapor (gas). This shows that the state of matter a substance is in can change under different conditions, even though the substance itself remains the same.

Particle Arrangement in the Three States of Matter

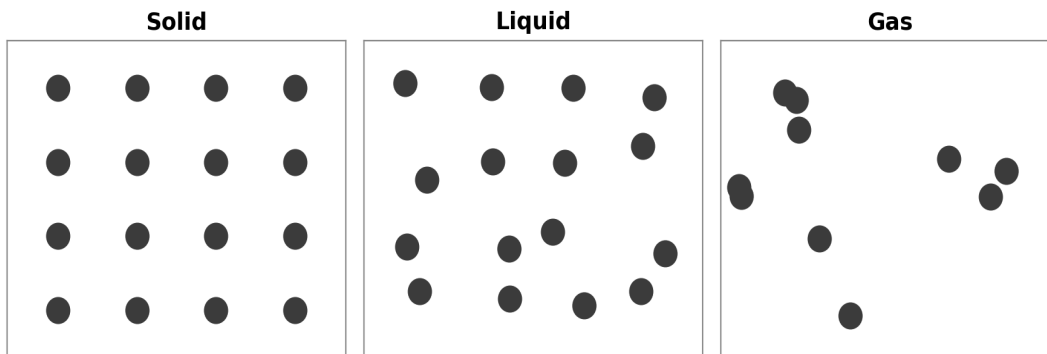


Figure 1. Particles are tightly packed and fixed in a solid, close together but able to move in a liquid, and spread far apart in a gas.